ModelIt!

Teacher's Guide

The Antibiotic Resistance Arms Race

How Superbugs Are Winning — And Why We Created Them

Quick Reference

Grade Level	Grade 09L2
NGSS Standard	HS-LS4-2, HS-LS4-3
Time	45–60 minutes (core lesson)
Materials	Student Activity Pack, devices with ModelIt! access
Prep	Print activity packs, test ModelIt! on student devices
Grouping	Partners or small groups (2–4)

ModelIt! — Discovery Collective

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1 NGSS Standards & CAST Alignment

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Performance Expectation: HS-LS4-2, HS-LS4-3

Construct an explanation based on evidence that the process of evolution primarily results from four factors: (1) the potential for a species to increase in number, (2) the heritable genetic variation of individuals in a species due to mutation and sexual reproduction, (3) competition for limited resources, and (4) the proliferation of those organisms that are better able to survive and reproduce in the environment. Apply concepts of statistics and probability to support explanations that organisms with an advantageous heritable trait tend to increase in proportion to organisms lacking this trait.

Three-Dimensional Learning

Science Practice	**Developing and Using Models** — Students develop a computational model with reinforcing feedback loops to explain how natural selection drives antibiotic resistance in bacterial populations under different treatment conditions.
Disciplinary Core Idea	**LS4.B Natural Selection / LS4.C Adaptation** — Natural selection acts on the variation in bacterial populations, where antibiotic exposure provides selective pressure favoring resistant individuals. Adaptation occurs at the population level as resistant traits become more prevalent.
Crosscutting Concept	**Cause and Effect: Mechanism and Prediction** — Students trace the causal chain from antibiotic misuse through selective pressure to resistance evolution, using their model to predict outcomes of different treatment strategies.

CAST Testing Alignment

MULTIPLE CHOICE:

A patient takes antibiotics for 3 days instead of the prescribed 10 days because they feel better. According to the reinforcing feedback loop model, what is the most likely long-term consequence?

CONSTRUCTED RESPONSE:

Using your computational model, explain the reinforcing feedback loop that drives antibiotic resistance. Include in your explanation why this loop makes resistance an accelerating problem rather than a self-correcting one, and propose an evidence-based intervention to slow the loop.

2 Scoring Rubric

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Assessment Reference

This is the same rubric students see in their Activity Pack. Use it during and after the lesson for formative and summative assessment. Circle or highlight the proficiency level for each skill.

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

Scoring Guidance

Expert (4): Student demonstrates deep understanding and can transfer knowledge to new contexts.
Proficient (3): Student meets grade-level expectations and shows solid understanding.
Developing (2): Student is making progress but needs additional scaffolding or practice.
Beginning (1): Student needs significant support; consider using Differentiation strategies (Section 7).

3 Background Content Knowledge

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For Your Confidence

This section provides the science background you need to facilitate this lesson with authority. You don't need to teach all of this — it's here so you can answer student questions accurately.

This lesson uses a feedback-dominant model to explore one of the most pressing threats to global health: antibiotic resistance. The core dynamic is a reinforcing feedback loop — as resistant bacteria increase in proportion, antibiotic effectiveness decreases, which allows even more resistant bacteria to survive. Unlike systems with balancing feedback that self-correct, this reinforcing loop drives exponential change, making resistance an accelerating crisis rather than a self-limiting problem. Students will explore how human behavior (dosage and duration choices) interacts with evolutionary mechanisms (mutation and natural selection) to produce outcomes that threaten modern medicine.

The Evolution of Resistance

Antibiotic resistance is evolution in real time. When Alexander Fleming discovered penicillin in 1928, he warned that bacteria would evolve resistance if the drug was misused. He was right. Bacteria reproduce every 20-30 minutes, with each generation introducing random mutations. Most mutations are neutral or harmful, but occasionally one confers resistance to an antibiotic. In a normal population, these resistant bacteria have no advantage. But when antibiotics are introduced, they kill susceptible bacteria while resistant ones survive and multiply — classic natural selection. The key insight is that antibiotics don't CAUSE resistance; they CREATE THE CONDITIONS where pre-existing resistance becomes advantageous.

The Reinforcing Feedback Loop

The dangerous dynamic in antibiotic resistance is a reinforcing (positive) feedback loop. As the proportion of resistant bacteria increases, the effective antibiotic concentration decreases (because fewer bacteria are susceptible). Lower effectiveness means more bacteria survive treatment, which further increases the resistant proportion. This loop accelerates over time — each cycle makes the next cycle worse. In systems thinking, this is called a 'vicious cycle' because it drives the system toward an increasingly undesirable state. The only way to break the loop is to interrupt it early, before resistance becomes dominant.

Why Incomplete Treatment Is Worse Than None

This is the counterintuitive insight that makes the model powerful: taking antibiotics for only a few days is MORE dangerous than not taking them at all. With no antibiotics, resistant and susceptible bacteria coexist without selective pressure favoring either group. But a partial course kills most susceptible bacteria while leaving resistant ones alive. The resistant bacteria, now freed from competition, multiply rapidly. The patient 'feels better' because the bacterial load dropped, but the remaining population is now dominated by resistant strains. Next time, the same antibiotic may not work at all.

The Global Crisis

The WHO has declared antibiotic resistance a top-10 global public health threat. In the US alone, over 2.8 million antibiotic-resistant infections occur annually, causing over 35,000 deaths. Some bacteria, like MRSA (methicillin-resistant Staphylococcus aureus) and CRE (carbapenem-resistant Enterobacteriaceae), resist nearly all existing antibiotics. The pipeline for new antibiotics is nearly empty because pharmaceutical companies find antibiotics less profitable than drugs for chronic conditions. Without effective antibiotics, routine surgeries, cancer chemotherapy, and organ transplants become life-threatening procedures.

4 The LEVER Framework

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About LEVER

LEVER is how every ModelIt! lesson works. Students move through five phases to build, test, and refine a computational model. Each lesson is self-contained — students complete all five phases in one session.

L

Locate the System

Students identify six components of the antibiotic resistance system: Antibiotic Dosage and Treatment Duration as external variables (human-controlled), and Bacterial Population, Resistant Bacteria Percentage, Mutation Rate, and Immune Response as internal variables (system-determined). Students must recognize that human choices directly drive the selective pressure that feeds the reinforcing loop.

E

Establish Relationships

Students map the reinforcing feedback loop: Antibiotic Dosage kills susceptible bacteria → Resistant Bacteria Percentage increases → Antibiotic effectiveness decreases → more resistant bacteria survive → Resistant Bacteria Percentage increases further. They also identify the balancing role of Immune Response and the threshold effect of Treatment Duration (completing vs. not completing the course).

V

Visualize & Model

Students build a multi-component model showing how Antibiotic Dosage and Treatment Duration interact with Bacterial Population dynamics, the reinforcing feedback loop driving Resistant Bacteria Percentage, the modulating effect of Mutation Rate, and the supporting role of Immune Response.

E

Evaluate Outcomes

Students run scenarios comparing full course, incomplete course, and sub-therapeutic dosage, evaluating which conditions activate the reinforcing feedback loop most dangerously. They analyze why incomplete treatment produces worse outcomes than no treatment — a counterintuitive prediction their model must explain.

R

Revise & Extend

Students extend the model by adding Horizontal Gene Transfer Rate, Antibiotic Diversity, or Biofilm Formation to explore how real-world complexity accelerates or mitigates resistance evolution beyond the basic reinforcing loop.

Dropping ModelIt! Into Your Existing Unit

ModelIt! is a **supplement**, not a replacement for your science curriculum. Each lesson stands alone, but it becomes even more powerful when students revisit their model throughout a unit. Here are easy ways to weave it in:

At the start of a unit	Use this lesson to introduce the phenomenon. Students build a “first draft” model based on what they already know. Save it.
After a reading or lab	Have students reopen their model and update it: “Based on what you just learned, would you add a component? Change a relationship?” Takes 10–15 minutes.
As a “what-if” tool	When students ask questions during class, say: “Great question — go test that in your model!” Students run new scenarios and report back.
For peer discussion	Partners compare models: “Why does your model have this arrow and mine doesn’t?” Differences spark scientific debate using evidence.
As a presentation tool	Students project their model and walk the class through it live — showing components, running scenarios, answering audience “what-if” challenges in real time.
At the end of a unit	Compare Day 1 model to final model. The growth IS the assessment — students see how their thinking evolved.

5 Slide-by-Slide Facilitation

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Slide 1: Cover

3 min

Say: *"Right now, bacteria are evolving to defeat every drug we have. And the terrifying part? We're helping them do it."*

Watch For

Show a news headline about a recent superbug outbreak. Ask: 'Has anyone heard the term superbug before? What do you think it means?'

Slide 2: Learning Objectives

4 min

Say: *"Today we're modeling one of the most dangerous feedback loops on Earth — one that's happening inside hospitals and medicine cabinets right now."*

Watch For

Have students read objectives. Pre-teach 'reinforcing feedback loop' using a simple everyday example (rumor spreading). Introduce vocabulary with emphasis on 'selective pressure' as the driving force.

Slide 3: Big Question

3 min

Say: *"Alexander Fleming, the man who discovered penicillin, warned us this would happen. In his 1945 Nobel Prize speech, he predicted that misuse of antibiotics would create resistant bacteria. He was right."*

Watch For

Share Fleming's actual Nobel Prize quote. Quick poll: 'Have you ever stopped taking antibiotics early because you felt better?' (Don't shame — this is about understanding why it matters.)

Slide 4: LEVER Framework

3 min

Say: *"We're building a model with a reinforcing feedback loop — a system that amplifies itself. Unlike self-correcting systems, this one spirals out of control."*

Watch For

Preview LEVER steps. Draw the reinforcing loop on the board: more resistance → less effectiveness → more resistance. Ask: 'What makes this loop dangerous compared to a balancing loop that self-corrects?'

Slide 5: Activity 1: Components

8 min

Say: *"We have six components. Two are things WE control — dosage and duration. Four are things the SYSTEM controls. Which is which, and why does that distinction matter?"*

Watch For

Guide component sorting. Key discussion: Why are Antibiotic Dosage and Treatment Duration external? Because humans decide them. Why is that important? Because the crisis is driven by human CHOICES, not just biology.

Slide 6: Activity 2: Connections

10 min

Say: *"Here's where it gets interesting. Trace what happens when you increase antibiotic dosage — what happens to bacterial population? And then what happens to resistant bacteria percentage? And then what happens to the antibiotic's effectiveness?"*

Watch For

Students map relationships. Explicitly identify the reinforcing loop. Have students trace the loop multiple times, noting how each cycle amplifies the previous one. Discuss: 'How is this different from the balancing feedback loops we've seen before?'

Slide 7: Activity 3: Simulation

12 min

Say: *"We're running three scenarios. One is proper treatment. One is what most people actually do. One is what happens when the dose is too low. Predict which is the MOST dangerous — then test your prediction."*

Watch For

Students predict outcomes BEFORE running simulations. Run all three scenarios. Key revelation: incomplete treatment produces WORSE long-term outcomes than no treatment. Have students explain WHY using the reinforcing feedback loop.

Slide 8: Discoveries

5 min

Say: *"Here's the takeaway that should genuinely scare you: 2.8 million antibiotic-resistant infections happen in the US every year. Over 35,000 people die. And the pipeline for new antibiotics is nearly empty."*

Watch For

Lead discussion connecting model results to real-world data. Key question: 'If your model shows that incomplete treatment accelerates resistance, what does that mean for the millions of people who stop antibiotics early every year?' Discuss systemic vs. individual responsibility.

Slide 9: STEM Challenge

12 min

Say: *"The WHO says antibiotic resistance is a top-10 global health threat. Your team has been hired to create a campaign that changes behavior — not just awareness. Your model data is your evidence."*

Watch For

Groups design evidence-based campaigns using model data. Emphasize that effective campaigns change BEHAVIOR, not just knowledge. Groups present and peer-evaluate campaigns on scientific accuracy, persuasiveness, and actionability.

6 Answer Key & Expected Responses

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Component Sorting

- **EXTERNAL:** Antibiotic Dosage, Treatment Duration
- **INTERNAL:** Bacterial Population, Resistant Bacteria Percentage, Mutation Rate, Immune Response

Relationships

- **Antibiotic Dosage to Bacterial Population:** NEGATIVE (-) — Higher antibiotic dosage kills more susceptible bacteria, reducing the total Bacterial Population. However, this effect diminishes as Resistant Bacteria Percentage increases — a critical non-linearity in the system.
- **Treatment Duration to Resistant Bacteria Percentage:** NEGATIVE (-) — Longer, complete treatment courses reduce Resistant Bacteria Percentage by ensuring that even partially resistant bacteria are eliminated. Incomplete treatment leaves resistant bacteria alive and removes their susceptible competitors.
- **Bacterial Population to Resistant Bacteria Percentage:** POSITIVE (+) [Reinforcing Loop] — As susceptible bacteria are killed by antibiotics, the Resistant Bacteria Percentage increases because resistant bacteria survive and reproduce without competition. This is the entry point to the reinforcing feedback loop.
- **Resistant Bacteria Percentage to Bacterial Population:** POSITIVE (+) [Reinforcing Loop] — As resistance increases, antibiotics become less effective, allowing the total Bacterial Population to rebound — now dominated by resistant strains.
- **Mutation Rate to Resistant Bacteria Percentage:** POSITIVE (+) — Higher mutation rates increase the probability of resistance genes appearing in the population, feeding the reinforcing loop with new resistant individuals.
- **Immune Response to Bacterial Population:** NEGATIVE (-) — A strong immune response works alongside antibiotics to reduce bacterial population. When immune response is weak, antibiotics must do all the work, increasing the chance of incomplete clearance.

Key Discoveries

- Antibiotics don't create resistant bacteria — they create the CONDITIONS for resistant bacteria to thrive by killing susceptible competitors
- Incomplete antibiotic courses are worse than no treatment at all because they selectively breed resistance
- The reinforcing feedback loop means once resistance passes a threshold, the same antibiotic becomes essentially useless
- Bacteria can share resistance genes through horizontal gene transfer, meaning one resistant species can arm others
- The mutation rate is constant, but selection pressure from antibiotics determines whether those mutations matter
- Immune response is a critical partner — antibiotics alone cannot eliminate infections if the immune system is compromised

7 Collaborative Learning Protocols

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Why This Matters

ModelIt! is a digital tool, but the deepest learning happens between humans. Computational modeling is most powerful when students build understanding *together* — debating relationships, challenging predictions, and constructing explanations as a community. These protocols ensure every student's voice is part of the model. Use at least ONE per lesson.

Model Gallery Walk

When: After students build models (LEVER Phase V)
Setup: Groups post their models on screens/desks. Half the class stays to "defend," half rotates with sticky notes.
Protocol: Visitors leave 2 sticky notes: one "I notice..." and one "I wonder..." After rotation, defenders read feedback aloud. Class discusses patterns.
Why it works: Students see that the SAME phenomenon can be modeled differently — and that's how real science works. (Bandura: learning through observation of peers)

Scientist Fishbowl

When: During simulation discussion (LEVER Phase E)
Setup: 4-5 students sit in center circle. Rest of class observes with clipboards.
Protocol: Inner circle discusses: "What happened in Scenario 3 and WHY?" using only evidence from their models. Outer circle records strongest arguments. Swap groups.
Why it works: Students practice scientific discourse without teacher directing. Observers learn to evaluate arguments. (Freire: dialogue as the foundation of critical consciousness)

Component Jigsaw

When: During component analysis (LEVER Phase L)
Setup: Assign each group ONE component to become "experts" on. They research it, discuss it, draw it.
Protocol: Regroup so each new team has one expert per component. Experts teach their component to the team. Team builds the model together using everyone's expertise.
Why it works: Every student is essential — the model literally cannot be built without each person's knowledge. Interdependence drives engagement. (Lave & Wenger: learning through legitimate peripheral participation)

Prediction Debate

When: Before running simulations (LEVER Phase E)
Setup: Post a scenario. Students physically move to corners: "Fire Spread will INCREASE a lot" / "INCREASE a little" / "STAY the same" / "DECREASE."
Protocol: Each corner group has 2 minutes to build their argument using model logic. One spokesperson presents. Then RUN the simulation. Losing corners explain what they missed.
Why it works: Movement + social risk + immediate feedback. Students CARE about the outcome because they committed publicly. Wrong predictions become the best learning moments.

Pair Programming for Models (Digital Collaboration)

When: Throughout model building
Setup: Two students, one device. **Driver** controls the mouse/keyboard. **Navigator** directs strategy and catches errors. Switch every 5 minutes.
Protocol: Navigator must verbalize every decision: "I think we should add a negative arrow from Rainfall to Dry Vegetation because..." Driver cannot act without Navigator's reasoning.
Why it works: Forces articulation of scientific thinking. The model becomes a shared artifact, not an individual assignment. Students who struggle with writing can demonstrate understanding verbally. Borrowed from computer science — where the best code is written in pairs.

Quick Protocol Selection Guide

Short on time (5 min)	Prediction Debate — gets bodies moving, generates energy fast
Full period available	Component Jigsaw — deep collaboration, every student contributes

After simulation day	Model Gallery Walk — peer feedback, multiple perspectives
Advanced discussion	Scientist Fishbowl — builds academic discourse skills
Every day default	Pair Programming — transforms individual work into dialogue

8 Differentiation & Extensions

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Below Grade Level

- Pre-sort components as a whole class
- Provide sentence frames for relationships
- Use visual vocabulary cards
- Pair with a strong reader

Advanced Learners

- Add 2-3 additional components
- Research real-world data
- Create multi-step causal chains
- Compare to similar systems globally

English Language Learners

- Pre-teach key vocabulary
- Provide bilingual glossary
- Use visual supports with text
- Allow verbal responses

Gifted & Talented

- Create feedback loop diagrams
- Research prevention strategies
- Write a policy brief
- Connect to climate change

9 Common Misconceptions

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Why This Matters

Addressing misconceptions proactively helps students build accurate scientific understanding. Use these as formative assessment checkpoints during the lesson.

STUDENTS MAY THINK	SCIENTIFIC UNDERSTANDING
"Antibiotics stop working because your body becomes immune to them"	It's not YOUR body that becomes resistant — it's the BACTERIA. Antibiotic resistance is an evolutionary adaptation of bacterial populations, not a change in human physiology. The bacteria evolve through natural selection: those with resistance genes survive treatment and pass those genes to their offspring.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Antibiotics create resistant bacteria"	Antibiotics don't CREATE resistance — they CREATE THE CONDITIONS where pre-existing resistance becomes advantageous. Resistance mutations exist randomly in bacterial populations before any antibiotic exposure. Antibiotics just remove the competition, allowing those rare resistant individuals to dominate.
STUDENTS MAY THINK	**SCIENTIFIC UNDERSTANDING**
"Taking more antibiotics is always better"	Higher doses and longer courses aren't always better — the optimal strategy is taking the RIGHT dose for the RIGHT duration as prescribed. Excessive antibiotic use (including in livestock agriculture) creates unnecessary selective pressure that breeds resistance without medical benefit. The goal is to use antibiotics precisely, not maximally.